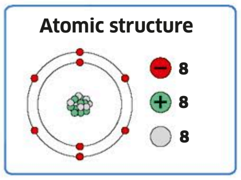
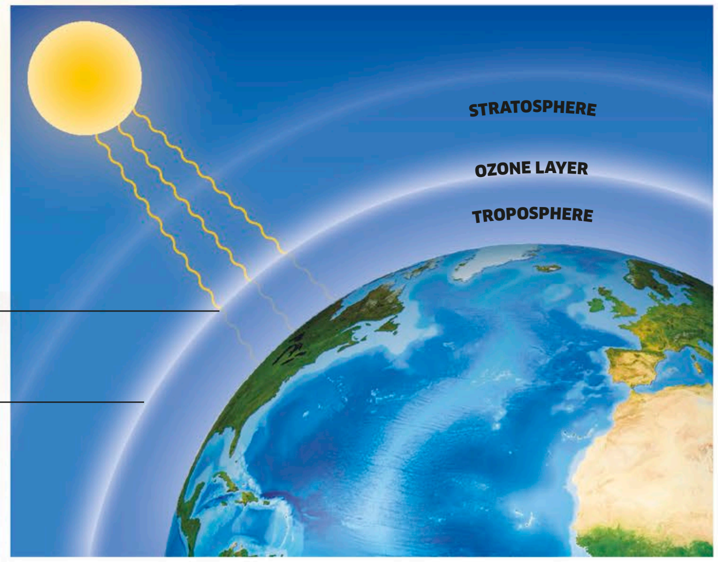
OXYGEN

Oxygenium

Discovered: 1774

The element that we depend on to stay alive, oxygen was only recognized as an element in the late 18th century. Many chemists from different countries had for years been trying to work out precisely what made wood burn, and what air was made of, and several came to similar conclusions at roughly the same time. Oxygen is useful to us in many different forms and roles, some of which are described here.





Most of the harmful UV radiation from the sun is absorbed by the ozone layer.

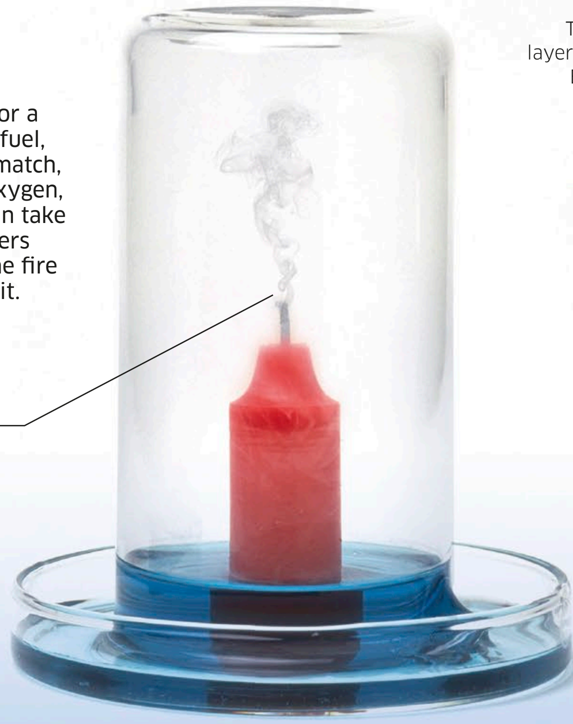
The ozone layer encircles Earth at a height of around 65,000 ft (20 km).

Air

Approximately 21 percent, or one fifth, of the air in Earth's atmosphere is oxygen gas. In the lower atmosphere, the oxygen we breathe is the most common form of oxygen—molecules made up of two oxygen atoms (O_2). Higher above us, however, is the ozone layer that protects us from harmful ultraviolet rays from the sun. Ozone (O_3) is another form, or allotrope, of oxygen, with three oxygen atoms in its molecules.

Fire

Three things are required for a fire to burn: there must be fuel, a source of heat such as a match, and oxygen gas. Without oxygen, no combustion (burning) can take place. Some fire extinguishers spray a layer of foam on the fire to prevent oxygen feeding it.

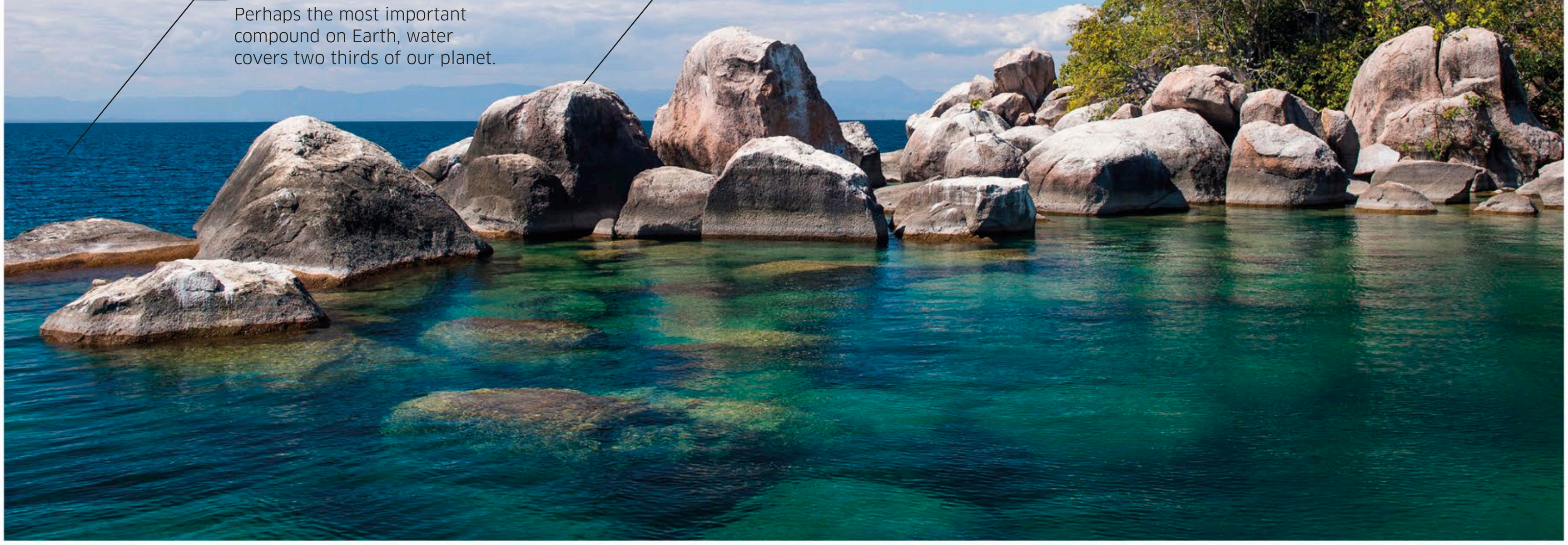


If a burning candle is placed in a jar, once the oxygen in the jar has been used up the flame soon flickers and goes out.

Life
All animals need oxygen to break down food and produce energy in a vital process called respiration.

Life on Earth

Our planet is the only one that has oxygen in its atmosphere. This is necessary for us to breathe. Oxygen is produced by photosynthesis, the process by which plants produce the food they need to live and grow. Water–which enabled life in the first place, millions of years ago, and is crucial to the survival of life in all forms–also contains oxygen. Even the ground is full of oxygen, in the form of different mineral compounds (see pp.22–23).



Land
Most of Earth's crust is made of rocks that contain oxygen compounds, such as these granite boulders.

Water
Perhaps the most important compound on Earth, water covers two thirds of our planet.